Real-Time Lane Detection and Autonomous Steering

A comparison of classical pipelines and a polynomial-fit improvement

David Tovmasyan

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Where we are going

Motivation

The simulator

Three lane detectors

Two controllers

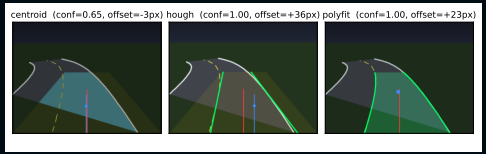
Experiments

Live demo

Discussion and conclusion

Why lane keeping (still) matters

- Most-deployed AD function: every modern car ships lane-keep assist or adaptive cruise.
- Two coupled problems
  - **Perception:** where is the lane?
  - **Control:** how do we stay in it?
- Learned segmenters dominate benchmarks, but a classical pipeline is:
  - interpretable (debuggable),
  - CPU-friendly (> 500 FPS),
  - zero training data required.



Same frame, three detectors.

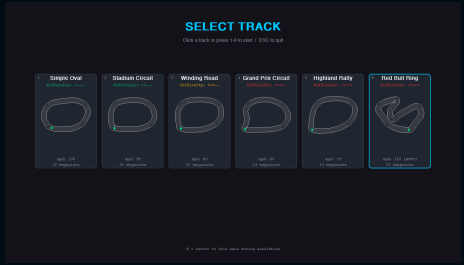
The concrete question

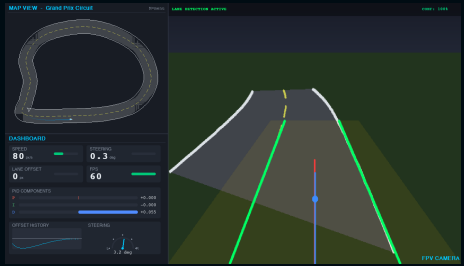
How much performance is left on the table
by the textbook Canny+Hough detector,
and can a **polynomial-fit upgrade**
close the gap on curved tracks
while staying **real-time**?

- Implement three classical methods of increasing sophistication.
- Plug each into a PID controller.
- Add a Stanley controller driven from *ground truth* as a perception-free oracle (upper bound on what the controller can do given perfect perception).
- Evaluate on five tracks + a noise sweep.

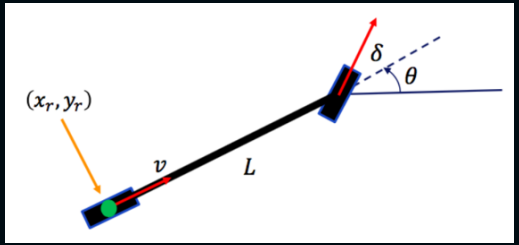
What we built

- 2D forward-perspective driving simulator (Python + OpenCV + pygame).
- 5 tracks (difficulty 1–5), closed-loop Catmull–Rom spline.
- Camera: pinhole, 70° FoV, 18° pitch, 640×480 FPV.
- Vehicle: bicycle kinematics, $\dot{\theta} = (v/L)\tan\delta$.
- Three live panels: map, FPV overlay, dashboard.





Vehicle model: bicycle kinematics



Rear-axle reference point (x_r, y_r), forward speed v, wheelbase L, heading θ, front-wheel steering angle δ.

Continuous-time kinematics integrated each frame:

$$\dot{x}_r = v\cos\theta$$

$$\dot{y}_r = v\sin\theta$$

$$\dot{\theta} = \frac{v}{L}\tan\delta$$

- Front-wheel steering, rear-axle reference – the standard simplification for lane-keeping work.
- δ clamped to $\pm 40°$; Δt capped at 50 ms to keep the Euler integration stable.
- No slip, no tire model – realistic enough for highway- speed lane keeping, deliberately simple so the comparison is about perception, not dynamics.

Ground truth = analytical

Because we own the simulator, we can compute the things a benchmark is normally missing:

- **True lateral offset** from the centreline spline.
- **True heading error** vs. the track tangent.
- **True curvature** ahead (adaptive speed control, turn-sharpness grouping).
- **Ground-truth lane polygon** in the camera frame (re-project road from spline) $\Rightarrow$ **lane-IoU** against any predicted mask.

$\Rightarrow$ Every metric is anchored to a number the detector *cannot see*.

Method 1: centroid (our baseline)

Pipeline

1. Threshold the grey road colour in HSV.
2. Mask to a trapezoidal ROI.
3. Centroid of the blob (image moments).
4. EMA smoothing across frames.

Role

- Simplest detector we could write $\Rightarrow$ adopted as the **reference baseline**.
- Tells us how much the lane-line geometry of Methods 2–3 actually buys.

Spoiler

Does not complete a lap on any track.

The blob centroid drifts off-centre whenever the road is asymmetric in the field of view.

Method 2: Hough lines (the textbook pipeline)

Pipeline

1. Blur, then Canny edges.
2. Trapezoidal ROI – keep only the road.
3. Probabilistic Hough $\Rightarrow$ line segments.
4. Split by slope sign: left vs. right lane.
5. Length-weighted average $\Rightarrow$ one line per side.
6. Lane centre = midpoint at the bottom.

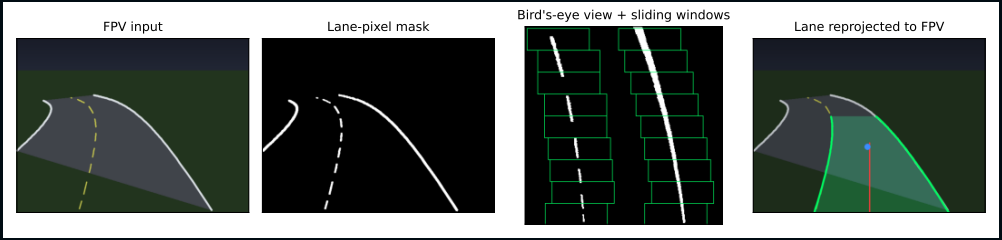
Strength

- Fast, no training, fully interpretable.
- Rock-solid on straight road.

Weakness

- **Models every lane as a straight line.**
- In a curve, the line cuts the corner – the error grows with the bend.

Method 3: bird's-eye + sliding-window polyfit



Four stages, each fixing a specific weakness of Method 2:

- **Lane pixels** – colour + edges, not just Canny.
- **Bird's-eye warp** – the model sees true road shape.
- **Sliding windows** – robust on dashed lines.
- **Quadratic fit** – a curve, not a straight line.

Method 3, step 1 – undo the perspective

Problem. In the camera image, the same physical dash covers ~ 30 px near the car but only ~ 2 px far ahead. A pixel-counting curve fit therefore listens to the near road and **ignores the far field** – exactly the part that announces upcoming curves.

Fix. A fixed homography warps the ROI trapezoid into a square top-down view. In that view:

- Every dash is the same size, near or far.
- Curved lanes become **gentle bends**, not vanishing-point lines.
- A 2nd-order polynomial $x = ay^2 + by + c$ fits a typical road curve almost exactly.

Why this is the key step

Without the warp, even a quadratic fit barely beats Hough, because the near road still dominates.

With the warp, near and far pixels get **equal say** in the fit.

Method 3, step 2 – find each lane

Inside the bird's-eye view, which pixels belong to the left vs. right lane?

Sliding-window search

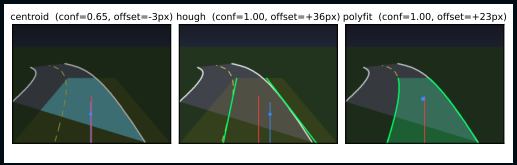
1. **Seed**: column histogram of the bottom half $\Rightarrow$ two strongest peaks = lane bases.
2. **Walk upward**: stack of 9 rectangular windows above each base; collect bright pixels inside.
3. **Recentre**: if enough pixels are found, slide the next window over to follow them. Robust to dashed lines and short gaps.

Quadratic fit + reprojection

- Least-squares fit $x = ay^2 + by + c$ on the collected pixels, one polynomial per lane.
- EMA on the coefficients $\Rightarrow$ stable across frames.
- Warp the curve back to the camera image for the overlay and for the lane-IoU metric.

Method 3, step 3 – the look-ahead offset

- The polynomial gives us the lane *shape*, not just one point on it.
- Evaluate the lane centre at **45 % up the BEV** – a 1-frame predictor, essentially free.
- Feed *that* to the PID instead of the bottom-of- image offset.
- Turn it off and the polyfit's RMS gain shrinks from $\sim 40\,\%$ to $\sim 10\,\%$.



The blue dot in the right panel is the reprojected look-ahead point.

PID – the baseline controller

$$\delta_t = K_p\, e_t + K_i \sum_{\tau \le t} e_\tau\, \Delta t + K_d\, \dot{e}_t$$

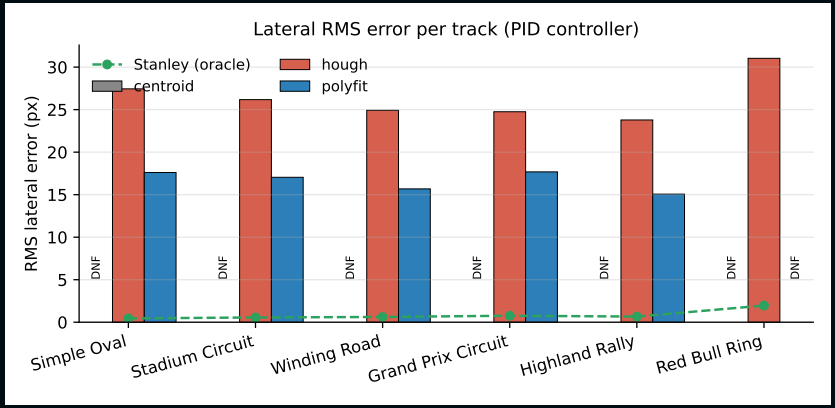
- Input: normalised lateral offset $e \in [-1, +1]$ from the detector.
- Gains tuned by hand on the oval: $K_p = 1.2$, $K_i = 0.008$, $K_d = 0.5$.
- Integral anti-windup at ± 50.
- Reactive only – has no model of the path.

Stanley – the perception-free oracle

$$\delta = \psi + \arctan\left(\frac{k\, e_\perp}{k_s + v}\right)$$

- Heading error ψ + cross-track error $e_\perp$ at the front axle, both computed from the *true* spline.
- Gains $k = 0.6$, $k_s = 20$.
- **It does not look at the camera.**
- Result on every track: < 1 px RMS lateral error.
- Interpretation: the gap between PID-on-detection and Stanley-on-truth is *purely* the perception error.

Headline result: RMS lateral error per track



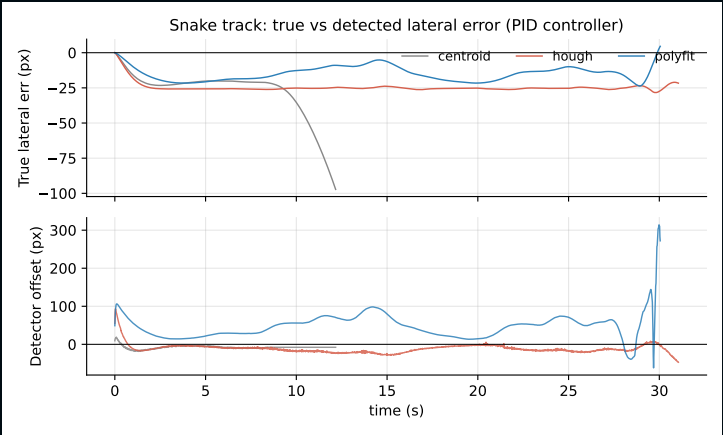
Centroid **DNF** on every track. On the 5 base tracks: Hough 25.4 px mean RMS, polyfit **16.6** px (**-34 %**). Red Bull Ring (hairpins, adaptive speed): **polyfit DNFs** at a hairpin, Hough survives at 31.0 px. Stanley+truth: < 2 px throughout.

The full table

Track	Detector	Lap	RMS px	Mean px	Max px	IoU	ms/det
Simple Oval	centroid	DNF	99.5	80.0	194.3	0.01	1.15
Simple Oval	hough	✓	27.4	27.2	28.7	0.17	0.75
Simple Oval	polyfit	✓	17.6	16.9	27.2	0.25	1.92
Stadium Circuit	centroid	DNF	39.5	32.8	113.4	0.39	1.79
Stadium Circuit	hough	✓	26.2	26.0	30.1	0.24	0.77
Stadium Circuit	polyfit	✓	17.0	16.3	23.6	0.31	1.94
Winding Road	centroid	DNF	33.9	28.2	97.8	0.41	1.84
Winding Road	hough	✓	24.9	24.7	28.4	0.23	0.76
Winding Road	polyfit	✓	15.7	14.8	23.7	0.35	1.96
Grand Prix Circuit	centroid	DNF	35.0	28.4	108.4	0.43	1.91
Grand Prix Circuit	hough	✓	24.8	24.6	27.8	0.26	0.77
Grand Prix Circuit	polyfit	✓	17.7	16.1	34.5	0.36	1.95
Highland Rally	centroid	DNF	65.6	52.9	127.6	0.00	1.22
Highland Rally	hough	✓	23.8	23.6	26.1	0.31	0.78
Highland Rally	polyfit	✓	15.1	14.2	21.9	0.39	1.95
Red Bull Ring	centroid	DNF	117.7	95.0	228.4	0.01	1.12
Red Bull Ring	hough	✓	31.0	30.5	51.9	0.21	0.73
Red Bull Ring	polyfit	DNF	28.5	22.8	82.0	0.24	1.90
Simple Oval	Stanley (oracle)	✓	0.4	0.4	1.2	0.52	2.04

- Where polyfit completes, it beats Hough on RMS and IoU ($-34\,\%$ RMS, $+38\,\%$ IoU averaged).
- **Red Bull Ring caveat**: the quadratic fit briefly blows up at a hairpin and the car leaves the road. Hough's straight line degrades gracefully and finishes.

Winding Road: where polyfit pays off

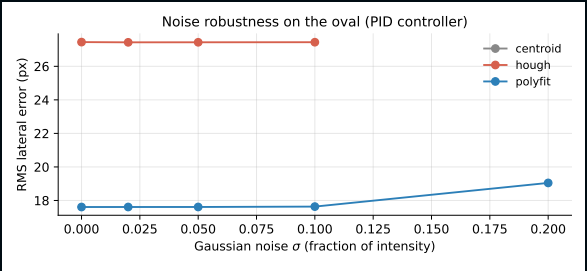


Top: true lateral error – centroid bails by $t \approx 12\,\text{s}$; Hough sits at $-25\,\text{px}$ through the S-curves; polyfit oscillates around 0 (follows bends, doesn't cut them).
Bottom: the detector offset fed to the PID – both polyfit and Hough stay well-behaved on this track.

Red Bull Ring: where polyfit falls over

Top: centroid bails at $t \approx 3$ s. Hough holds ~ 30 px to the flag. Polyfit trails Hough until $t \approx 17$ s.
Bottom: a hairpin makes the quadratic fit explode to ~ 8000 px for two frames – PID over-corrects, car off-track. **Fix:** clamp detector offset, or reject fit outliers.

Robustness to camera noise



- Same oval, PID, +Gaussian noise $\sigma \in [0, 0.20]$ per frame.
- Centroid never completes.
- Hough survives $\sigma \leq 0.10$, fails at 0.20 (Canny drowned out).
- **Polyfit completes at every σ** – its mask uses a *normalised* Sobel-x + colour.

Compute cost

Detector	ms / frame	equiv. FPS
centroid	1.7	$\sim$600
hough	0.7	$\sim$1450
polyfit	1.9	$\sim$520

- All three are comfortably real-time on a single CPU thread.
- Polyfit overhead is dominated by the perspective warp and `np.polyfit`, not by the sliding-window scan.
- Plenty of budget left for downstream tasks (other vehicles, obstacles, signs).

Live demo

```
python main.py --detector polyfit --controller pid
```

- Press **D** to cycle detectors live.
- Press **C** to toggle PID $\leftrightarrow$ Stanley.
- Press **M** to jump back to the track selector.
- Watch the dashboard: lane offset, FPS, PID components.

Suggested demo route: *Stadium* (Hough wobbles) $\rightarrow$ *Winding Road* (Hough struggles) $\rightarrow$ switch to polyfit $\rightarrow$ switch to Stanley.

What worked, what failed

Worked

- Look-ahead offset = single biggest gain.
- HSV + Sobel mask cleaner than Canny on dashed lines.
- Quadratic fits + EMA give stable curves.

Failed / limitations

- Centroid is structurally biased – no tuning saves it.
- Hough lingers on its last good line through tight bends.
- Polyfit's quadratic fit can **blow up at tight hairpins** (Red Bull Ring) – needs outlier rejection.
- IPM warp assumes flat ground; real cameras need re-tuning.
- Simulator is too clean – numbers are an upper bound.

Take-aways

- A modest classical-CV upgrade (BEV warp + sliding-window quadratic fit) cuts RMS lateral error by **34–40 %** vs. the textbook Hough pipeline.
- The change is fully interpretable: we can point at each stage in Fig. 3 and say why it helps.
- Look-ahead offsets are essentially free on top of a polynomial fit and turn a reactive PID into a quasi-MPC.
- Closing the remaining gap to Stanley-on-truth would need sub-pixel-accurate perception $\Rightarrow$ a learned segmenter is the next step.

Thank you.

Code, simulator, report and these slides are all in this repo.
Questions?

Backup: detector confidence per track

- Centroid mean confidence: **0.27** (HSV blob coverage, near zero whenever the road leaves the trapezoid).
- Hough mean confidence: **0.99** (both lines almost always found – but "found" $\neq$ "correct").
- Polyfit mean confidence: **1.00** (both polynomial fits valid in every frame after warm-up).
- Take-away: high detector confidence does not imply low error. Lane-IoU is the better proxy.

Backup: why the look-ahead works

Recall the bicycle heading update:

$$\dot{\theta} = \frac{v}{L} \tan \delta$$

- A pure-P controller on e_{bottom} reacts to *current* position only.
- Adding the look-ahead error e_{LA} effectively augments the controller with a free derivative term that knows where the road is going.
- Equivalent to a one-step preview MPC with horizon $\sim 0.5\,\text{s}$ at typical speeds – without solving an optimisation.

Backup: reproducibility

```
git clone <repo>
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt matplotlib pdf2image
python experiments.py   # CSVs + summary figures
python make_qualitative_figures.py   # overlays
cd presentation && tectonic report.tex
cd presentation && tectonic slides.tex
```

All experiments are deterministic except the noise sweep, seeded by NumPy's default RNG.